

# BiasLens: Graph Machine Learning Approach for Media Bias Detection

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**Abstract**—Bias in political media poses a serious concern in an age revolving around media consumption, as it shapes public perceptions and sows social discourse. Most bias detection methods focus solely on the textual content of an article, often overlooking the critical role of authors and news outlets in passing on their biases to articles. In this paper, we introduce BiasLens, a novel framework that leverages Knowledge Graphs (KGs) and graph embeddings incorporating different influences in bias classification. By constructing a KG that captures relationships among articles, authors, and publishers, we can enable a more context-aware bias detection approach. We used a labeled set of political news to construct a KG, applied structured graph-walking strategies, and extracted article node embeddings to train a classifier model. Our approach outperforms traditional text-based methods, achieving an accuracy of 96%, compared to 84% using a BERT-based classifier. These results highlight the potential that KG-based methods have in bias detection tasks and can lay the groundwork for future advancements in media bias detection.

**Index Terms**—Knowledge Graph, Media Bias Detection, Machine Learning.

## I. INTRODUCTION

The role of digital media cannot be understated, as many readers receive their news through these platforms and formulate their opinions and views on certain issues based on what they read. Media bias also has a significant role in this cycle as it can influence individuals' beliefs by reinforcing existing views through selective exposure [1], where people consume content that aligns with their pre-existing opinions, thus contributing to political polarization. Additionally, the hostile media effect [2] leads individuals with strong political views to perceive media coverage as biased against their position, further intensifying polarization. Media bias can have serious consequences for democratic discourse, as it erodes trust in media and democratic institutions. As a result, a need has emerged for techniques for spotting bias in media.

Traditional bias detection methods often rely on Natural Language Processing (NLP) techniques that analyze textual content. However, these approaches face challenges, including the potential for NLP algorithms to perpetuate existing biases present in their training data. In addition, such methods may overlook the influence an author or a publisher can have on

a given article, which we believe is important to account for.

We propose a graph-based approach that models the relationships between articles, authors, and publishers to address these limitations. By constructing a Knowledge Graph (KG), we aim to capture the contextual information that traditional text-based methods might miss. This approach allows for a more comprehensive understanding of media bias by taking into account the interconnected nature of news and media production.

In this paper, we introduce **BiasLens**, a novel framework that leverages KGs and graph embeddings for bias detection. Our key contributions include the construction of a media bias KG that encapsulates the relationships among news articles, their authors, and publishing outlets and the implementation of edge weighting strategies to enhance the KG's representational power and reflect the strength and influence of relationships. In addition, our approach is benchmarked against a BERT-based [3] model, demonstrating significant improvements in accuracy. Through BiasLens, we aim to provide a more nuanced and context-aware tool for media bias detection, contributing to promoting media transparency and informed public discourse.

The paper is structured as follows: Section II presents the literature review. Section III provides details on the dataset. Section IV outlines our baseline methodology for bias classification. Section V describes our approach to enhancing the knowledge graph with edge weighting. Section VI explains the embedding extraction process. Section VII presents our findings, and finally, Section VIII summarizes the results and discusses future work.

## II. RELATED WORKS

During our extensive literature review, we came across many examples of KGs being used in various recommendation and detection tasks similar to our approach, though to the best of our knowledge, using KGs to detect news bias is a novel idea. Still, the use of KGs in the news domain, in general, is not new.

The current state-of-the-art method in bias detection involves centers around augmenting the input data fed to pre-trained models. [4] proposed Bias-Aware Neighborhood Contexts (BANC), this approach labels bias on a sentence level but considers the bias label of neighboring sentences and adds them to the input of the model. They tested their approach with models such as BERT [3] and RoBERTa [5] and found that the BANC method significantly improved F1-scores 58.15 with BERT, a notable improvement over the regular setup, which had an F1-score of 42.32 though is on a sentence level and not on article level. [6] proposed a KG approach for informing Fake News Detection using LLMs. Their methodology extracts graph embeddings from the WikiData5M KG based on the most relevant triplets in an article text. Combining these embeddings with the vector obtained from embedding the article corpus using LLMs resulted in a Knowledge-enriched document vector. Their enhanced document vectors outperformed their baseline with a maximum accuracy of 97.06% This approach caught our eye as it implements KGs in a task not too far off from our own.

Using KGs to enhance a news recommendation algorithm was discussed in [7] since the time-sensitive nature of news means that interaction data may be sparse and not fully representative of a user's use. Their solution to this problem was building a news-specific KG to fill the data sparsity of interaction data to recommend news more accurately, they managed to outperform other methods with an accuracy of 0.8503 and F1-score of 0.7992 in local news detection. Another task is fake news detection. [8] discussed the spread of news on social media and proposed DEAP-FAKED, a dual-part KG-based framework to solve the problem. Their approach combines NLP methods to encode news headlines and embedded KG entities. DEAP-FAKED achieved an F1-score of 88% and 78% on two public datasets, which outperformed other classification methods. In their work, [9] developed CompareNet, a graph neural model that differentiates between satirical and trusted news articles. CompareNet outperformed the state-of-the-art models, attaining 3% improvements in both micro and macro F1 scores. Their model benefited from topic information and connections to an external knowledge base (KB), but removing topic information led to significant performance decreases of 4.0-5.0% in F1 scores. This emphasizes the need to construct a robust and descriptive KG to fulfill the goal of our product.

While the aforementioned works demonstrate the potential of Knowledge Graphs (KGs) and graph-based embeddings in tasks like fake news detection and recommendation, they are generally not designed for article-level bias detection, which is the primary focus of our study. For instance, methods like Bias-Aware Neighborhood Contexts (BANC) and DEAP-FAKED are focused on sentence-level bias identification or fake news detection, which overlooks the broader structural

and contextual relationships between articles, authors, and publishers that shape the biases at the article level. While these approaches improve sentence-level accuracy, they do not capture the interconnectedness between entities that influence the overall bias of a news article. Similarly, techniques like CompareNet and NewsGraph, though beneficial in their respective domains, focus primarily on improving news recommendation systems or distinguishing between satirical and trustworthy content rather than detecting the political or ideological bias within individual articles. In contrast, our BiasLens framework is explicitly designed to model and analyze the intricate relationships between articles, authors, and publishers, allowing for a more context-aware approach to bias detection at the article level. This difference underscores the unique contribution of our work in advancing media bias detection by incorporating these broader contextual relationships into the model, which can lead to more nuanced and accurate results at the article level.

### III. DATASET AND KNOWLEDGE GRAPH CONSTRUCTION

The dataset used in this study is by [10]. It consists of 37,737 articles collected from AllSides, each article is labeled with a political bias rating of either left, center, or right. The class distribution is shown in figure 1. It is important to mention that the authors state that "center" means a centrist political view rather than the lack of a bias or "no bias". This dataset stands out as AllSides employs a systematic process with various steps including a Blind Bias Survey<sup>1</sup> where individuals with various political ideologies read headlines and provide a bias rating without knowing the source the article comes from which both saves time and eliminates bias from our end if we were to label the dataset from our end.

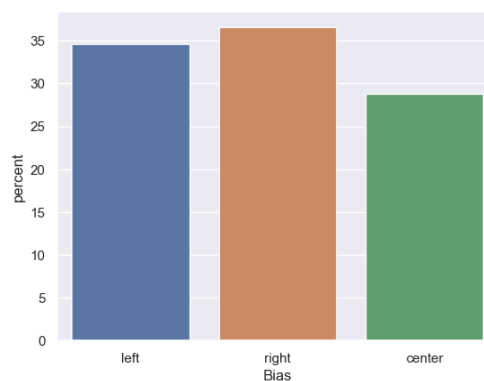


Fig. 1. Dataset Class Distribution

The dataset included detailed descriptions of each article, including the Title, Content, Topic, Source, Authors, and Bias. Using regular expressions (regex), we performed text

<sup>1</sup><https://www.allsides.com/blind-bias-survey>

cleaning on creator names since multiple authors can be associated with a single article separated by commas. We split the names and removed special characters and double spaces to make sure they are consistent across records. For missing names, we opted to replace them with "Anonymous" in order to preserve completeness and consistency when it comes time to construct the graph. In addition, we will stick with the three bias classes included in the dataset, thus, we observed the class distribution to make sure no class is over-represented. We do not intend to extract any features from the content of the article thus, we did not process their text.

The construction of KG was highly dependent on the structure of the dataset. Our methodology for constructing the KG can be split into 2 parts: The T-Box and the A-Box.

#### A. T-Box

From the available data, we chose to define the following classes: Article, Author, Outlet, Topic, and mentionedEntity. The classes can then be linked using the following relationships: "writtenBy" links an article with an author, and "publishedBy" links an article with an outlet, "hasTopic" links an article to its topic, "mentions" links an article to a mentioned entity. We chose these classes and relationships with future expansion in mind and included data properties for the article class to include: body, id, and most importantly, bias.

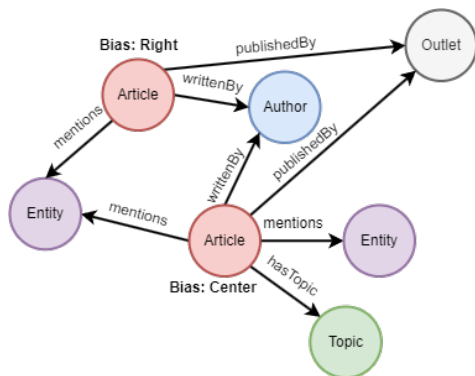


Fig. 2. Example Knowledge Graph Structure

#### B. A-Box

Using rdflib [11] python package, we iterated over the records in our dataset and mapped the individual records to instances of the article, author, and outlet classes. For each record, we created an Article instance with properties body, id, and bias filled with the corresponding data from the record. We also created Author and Outlet instances based on the author and outlet information in the record. For the mentioned entities, we employed a fine-tuned version of GLiNER [12]

to perform named entity recognition over each article and extract 3 main types of entities: a location, a person, or an organization. The extracted entities have to be disambiguated so we don't end up with multiple entities referring to the same thing "e.g. US and United States are the same entity", for that we can query the Wikidata KG [13] with the extracted entity and use the response as our final entity name.

### IV. ESTABLISHING A BASELINE FOR BIAS CLASSIFICATION

To evaluate the effectiveness of our graph-based approach, we first established a baseline model using a BERT-based sequence classification model. This baseline serves as a text-only bias classification approach, allowing us to compare traditional NLP methods with our Knowledge Graph (KG) approach.

#### A. Data Preprocessing

The dataset was cleaned and tokenized using a BERT tokenizer to ensure consistency in text representation. We removed special characters and unnecessary whitespace in addition to numbers, punctuations, and stopwords. At this step, we observed that many articles exceed the maximum sequence length supported by the base BERT model, averaging 1219.03 tokens per article. We, therefore, opted to use Longformer [14], a variant of BERT architecture that supports a sequence length of up to 4096 tokens. The dataset was split into training and testing sets, with 80% of the data used for training and 20% for testing.

#### B. Experiment Setup

We utilized Longformer for sequence classification, fine-tuning the base model on the dataset. The model was trained for a maximum of 10 epochs, with early stopping enabled to prevent over-fitting. The training setup was configured with AdamW Optimizer, a learning rate of  $2e-5$ , and Cross-Entropy Loss. Early stopping was set up to stop training if validation loss does not improve for 2 consecutive epochs.

#### C. Training Results

The model stopped training at epoch 5, attaining a classification accuracy of 0.84 on the test set.

TABLE I  
BERT PERFORMANCE METRICS PER CLASS

| Accuracy | Class-wise Metrics |           |          |
|----------|--------------------|-----------|----------|
|          | Recall             | Precision | F1-score |
| 0.84     | 0.89               | 0.79      | 0.84     |
|          | 0.83               | 0.85      | 0.84     |
|          | 0.81               | 0.88      | 0.85     |

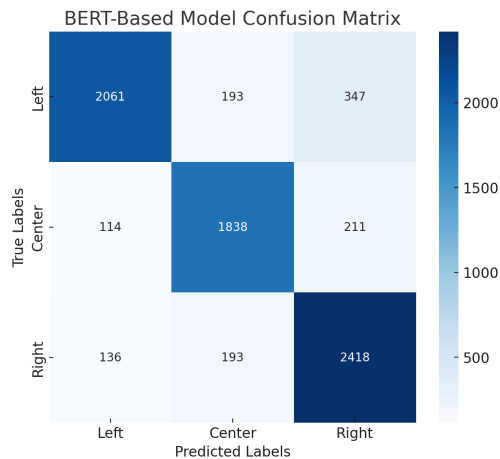


Fig. 3. BERT-Based Model Confusion Matrix

#### D. Limitations of the Text-Based Approach

While the BERT-based classifier demonstrated strong performance, it is inherently limited by its dependence on the textual content of the article without accounting for contextual relationships, such as author or publisher influence. This highlights the need for a more context-aware approach, motivating the shift to a KG-based bias detection framework, which captures structural relationships between news articles, authors, and publishers. The following sections explore how graph-based embeddings can address these shortcomings and enhance classification performance.

#### V. ENHANCING THE KNOWLEDGE GRAPH WITH EDGE WEIGHTS

Building further on the KG we implemented edge weighting to capture the varying strengths and influences of different relationships. The rationale for edge weighting is the ability to quantify the significance or strength of different relationships. For example, an author who frequently writes for a particular news outlet or on a specific topic may have a stronger influence on the articles' content, which can be represented by a higher edge weight. Incorporating such weights enables the model to differentiate between varying degrees of influence, leading to more precise bias detection. In pyRDF2Vec, different weighting strategies can be implemented with the use of different samplers, which change the priorities of walks generated for specific nodes, which in turn can change how those nodes' embeddings are computed. Table II summarizes the different sampling strategies used.

#### VI. GRAPH EMBEDDING METHODOLOGY

We employ the Python implementation pyRDF2Vec [15] of the RDF2Vec algorithm [16], which adapts the principles of the Word2Vec model [17] to the domain of KGs. This is achieved by generating random graph walks from a starting entity, treating said walks as sentences, and then feeding those sentences into the Word2Vec model to learn embeddings that

TABLE II  
SUMMARY OF DIFFERENT PYRDF2VEC SAMPLING STRATEGIES

| Sampler         | Description                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------|
| ObjFreqSampler  | Prioritizes walks containing edges with the highest degree objects.                         |
| UniformSampler  | Assigns a uniform weight to each edge in a Knowledge Graph.                                 |
| PageRankSampler | Prioritizes walks containing the most frequent objects using the PageRank ranking.          |
| WideSampler     | Gives priority to walks containing edges with the highest degree of predicates and objects. |
| PredFreqSampler | Prioritizes walks containing edges with the highest degree predicates.                      |

capture the semantic relationships between entities. Mathematically, the objective is to maximize the likelihood of observing a context entity  $c$  given a target entity  $t$ :

$$\max_{(t,c) \in D} \prod P(c | t) \quad (1)$$

where  $D$  represents the set of all (target, context) pairs derived from the generated sentences. Different experiments can be implemented, which can affect the "sentences" passed to Word2Vec, which in turn may result in different embeddings. Table III summarizes the walking strategies we will experiment with.

TABLE III  
SUMMARY OF WALK STRATEGIES

| Walker        | Description                                                                                                                                            |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| HALKWalker    | Removes rare vertices from randomly extracted walks.                                                                                                   |
| NGramWalker   | Relabels the n-grams in random walks to define a mapping from one-to-many.                                                                             |
| RandomWalker  | Extracts random walks from a root node using Depth First Search (DFS) if a maximum number of walks is specified, else, use Breadth First Search (BFS). |
| WalkletWalker | Transforms randomly extracted walks into walklets, which are walks of size one or two.                                                                 |

#### VII. MACHINE LEARNING FOR BIAS CLASSIFICATION

##### A. Data Processing

We generated a KG using rdflib [11], as previously stated in section III. We also keep a list of URI and bias label pairs in order to split the graph embeddings into training and testing sets.

##### B. Embedding Extraction

We follow section ?? and VI by utilizing different sampling and walking strategies in the pyRDF2Vec package [15]. We choose a test size of 20%, we also fix the training of the embedding model to 20 epochs. The key advantage of using pyRDF2Vec is the inclusion of a suite of walking and sampling strategies, we will use every combination of samplers and

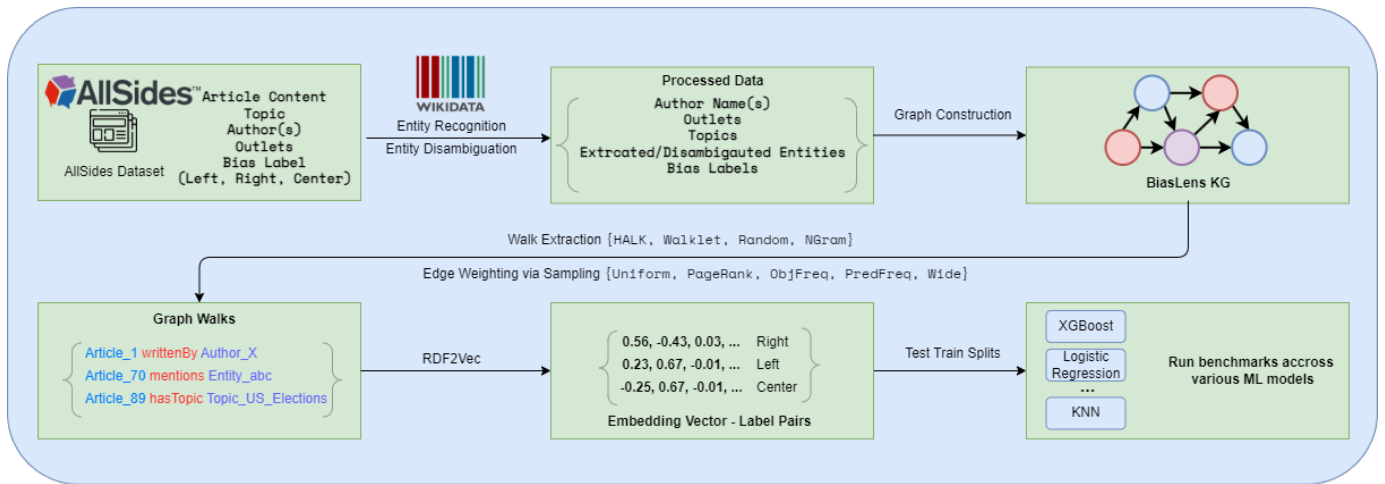


Fig. 4. BiasLens System Overview

walkers in the tables II and III and feed each combination's embeddings into each of the following models, XGBoost, Logistic Regression, Random Forest Classifier, SVM and KNN and score them based on accuracy and F1 metrics.

### C. Results

The results were more than promising, as Walklet Walker with PageRank sampling achieved the highest accuracy using the XGBoost Model, while Walklet Walker with Wide sampling achieved the highest F1 score using the same model. Comparing the results to the baseline, we find that both combinations outperform the baseline results presented in Table I. The results prove the efficacy of the BiasLens framework in enhancing the performance of bias detection tasks. Results for all combinations are shown in tables IV and V.

TABLE IV  
F1-MACRO SCORES OF WALKER-SAMPLER COMBINATIONS

| Walker / Sampler | ObjFreq | PageRank | PredFreq | Uniform | Wide          |
|------------------|---------|----------|----------|---------|---------------|
| HALKWalker       | 0.8968  | 0.8996   | 0.8997   | 0.8981  | 0.8938        |
| NGramWalker      | 0.9196  | 0.9139   | 0.9226   | 0.9169  | 0.9154        |
| RandomWalker     | 0.8983  | 0.8946   | 0.8894   | 0.89    | 0.898         |
| WalkletWalker    | 0.9415  | 0.9439   | 0.9429   | 0.9394  | <b>0.9439</b> |

TABLE V  
ACCURACY SCORES OF WALKER-SAMPLER COMBINATIONS

| Walker / Sampler | ObjFreq | PageRank      | PredFreq | Uniform | Wide   |
|------------------|---------|---------------|----------|---------|--------|
| HALKWalker       | 0.8977  | 0.9007        | 0.9004   | 0.8991  | 0.8951 |
| NGramWalker      | 0.9197  | 0.9144        | 0.9232   | 0.9173  | 0.9157 |
| RandomWalker     | 0.8993  | 0.896         | 0.8905   | 0.8915  | 0.8992 |
| WalkletWalker    | 0.9423  | <b>0.9447</b> | 0.9437   | 0.9403  | 0.9447 |

## VIII. DISCUSSION AND FUTURE WORK

In this work, we explored several sampling and walking strategies for generating graph embeddings through the pyRDF2Vec implementation. Each strategy influences

the learning of node embeddings in different ways, and understanding these differences is crucial for fine-tuning the bias detection model.

Walklet stands out among other graph walk strategies due to its ability to balance both local and global structural information. Unlike random walks, which may only capture immediate neighbor relationships, Walklet incorporates a memory-based approach, retaining past node visits to capture long-range dependencies. This is in contrast to simpler methods like random walks or N-gram sampling, which primarily focus on local node sequences without considering the broader context. Additionally, Walklet improves on samplers like PageRank by not only emphasizing node importance but also accounting for hierarchical, multi-scale relationships within the graph, which is reflected in the results.

PageRank and Wide sampling differ in their approaches to graph traversal. PageRank sampling prioritizes nodes based on their importance in the graph, selecting high-degree or influential nodes more frequently. These nodes could be the authors or outlets that often lean toward a specific bias. Wide sampling aims for diverse coverage, ensuring that nodes from different regions of the graph are sampled. This helps capture a broader variety of relationships for each article node, which in turn results in richer embeddings.

Furthermore, in our experiments, we utilized an embedding dimension of 200, a size commonly used in [18] across diverse machine learning tasks. Although the authors do not explicitly recommend a universal optimal dimension, their experiments frequently employed embedding sizes of 200, showing effective performance across different prediction scenarios. Given this precedence and the use of classifiers such as XGBoost and Random Forest, which typically perform well with embeddings in this dimensional range, selecting a dimension of 200 appeared logically



sound and computationally reasonable for our study. Explicit dimensionality tuning was not performed due to scope constraints, and it remains an open avenue for future exploration to potentially further optimize BiasLens performance.

A notable limitation of the current RDF2Vec methodology is its static nature. Traditional RDF2Vec embeddings are designed for fixed KGs, necessitating a complete retraining of the model upon any update to the graph. This retraining process can be computationally intensive and time-consuming, especially as the KG scales. To address this, recent advancements propose incremental updating methods for RDF2Vec embeddings, allowing for efficient updates without full retraining [19]. Implementing such techniques could enhance the adaptability of BiasLens in dynamic news environments.

Although our approach demonstrates strong performance, it is important to acknowledge certain limitations introduced by the dataset used. The dataset, collected by [10] from AllSides, relies on political bias collected from crowd-sourced Blind Bias Surveys. While this methodology helps mitigate explicit labeling biases, it may still indirectly incorporate subjective or demographic biases from survey participants. In addition, the dataset is confined to U.S. political news, which may affect the generalizability of our model to other types of media bias in different news domains. Future work should focus on validating BiasLens with datasets representing diverse regional contexts, multiple languages, varied forms of media content, and possibly a temporal aspect since it is not uncommon for news stories to be rapidly updated and for biases to shift as a news story develops. Addressing these limitations and iterating can help BiasLens to become the best version of itself.

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